

Astrodynamics I: Midterm Exam

April 26, 2011

1. If v_a and v_p are the apoapsis and periapsis speeds of an elliptic orbit, respectively, with semimajor axis a and eccentricity e , show that

A. Their arithmetic mean is $\frac{1}{2}(v_a + v_p) = \sqrt{\frac{\mu}{a(1-e^2)}}$

B. Their geometric mean is $\sqrt{v_a v_p} = \sqrt{\frac{\mu}{a}}$

2. A spacecraft is orbiting a planet in a circular orbit with a radius of 4 Planet Radii ($a=4, e=0$). A maneuver is performed that rotates the velocity vector in the orbital plane by an angle of 45° towards the planet.

A. Find the new semimajor axis a_n and the new eccentricity e_n following this maneuver.

B. At what point in its new orbit is the spacecraft in terms of true anomaly (ν)?

C. Will the spacecraft impact the planet? Justify your answer.

3. Starting from $x=r\cos(\nu)$ and $y=r\sin(\nu)$, show that

A. $\dot{x} = -\sqrt{\frac{\mu}{p}}\sin\nu$ and $\dot{y} = \sqrt{\frac{\mu}{p}}(e + \cos\nu)$

B. $f \stackrel{\text{def}}{=} \frac{x\dot{y}_0 - \dot{x}_0 y}{h} = 1 - \frac{r}{p}(1 - \cos(\Delta\nu))$, $\Delta\nu = \nu - \nu_0$

Hint: $\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta)$

4. Given $\vec{r} = -\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ and $\vec{v} = \frac{1}{2}\hat{j}$, determine the orbital elements and sketch the orbit.

Use $\mu=L$.

Hint: $\vec{e} = \frac{(v^2 - \frac{\mu}{r})\vec{r} - (\vec{r} \cdot \vec{v})\vec{v}}{\mu}$